

Muhammad Talha Arshad

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Technical Skills

Programming: Python, R, C++, C#, Node.js

Machine Learning & AI: Statistical Analysis, Probability, ARIMA, SVM, Random Forest, Deep Learning, NLP, Neural Networks, Diffusion Models

Databases: SQL, MySQL

Tools: ERD, MER, Relational Mapping, LaTeX, Tableau

Languages: English, Urdu

Projects

Automated Multi-Source Climate Resilience & Socio-Economic Intelligence Pipeline (First Major Project)

Python — Pandas — NumPy — Tableau

- Engineered a Python ETL pipeline integrating 5+ global datasets (UNDP, World Bank, WGI, EM-DAT) with 50,000+ records across 200+ countries.
- Applied wide-to-long pivoting and cross-indicator merging to align multi-decade time-series datasets.
- Designed 15+ analytical indicators including Composite Resilience Index (CRI), Vulnerability, Exposure, and Disaster Impact Index.

Inflation Prediction Model — R, ARIMA, SVM, Random Forest

- Developed a predictive model for inflation forecasting using time-series and machine learning techniques.

Traffic Manager Simulation — C++

- Simulated traffic flow to analyze congestion patterns and evaluate traffic efficiency.

Game Information Database — C++

- Designed and implemented a structured database system for storing and managing game data.

Work Experience

Machine Learning Research Intern

Jun 2025 – Aug 2025

Knowledge Discovery & Data Science (KDD) Lab, FAST-NUCES

- Applied machine learning, NLP, and deep learning techniques to improve image generation workflows.
- Developed models trained on multilingual prompt components using scene graph representations.

Volunteer Data Analyst

Feb 2025 – May 2025 Jamshed Akhtar Qureshi Education Trust (JAQET)

- Processed and cleaned 100+ survey datasets including normalization and preparation.
- Conducted statistical analysis using Python to support data-driven educational planning.

Education

Bachelor of Science in Data Science

FAST – National University of Computer and Emerging Sciences (FAST-NUCES)

2023 – 2027

Relevant Coursework: Machine Learning, Statistical Analysis, Data Mining, Database Management, Data Warehouse, Deep Learning, NLP, AI, Advanced Statistics, Data Visualization

Certifications & Achievements

Supervised Machine Learning — Stanford University

Probability for Data Science — Great Learning Academy

Data Science Tools — IBM Cognitive Class

Scholarship Holder (10th Grade) — FBISE